Jeslyn Wang

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EDUCATION

B.A.Sc in Computer Engineering + PEY Co-op, University of Toronto Expected June, 2026

- CGPA: 3.78, Dean's Honour List for 6 terms
- Intended minors in Artificial Intelligence and Engineering Business
- GenAI Hackathon 2025 Organizer, Marketing Chair for Canada's largest AI hackathon

SKILLS

- **Languages:** Python, C, C++, C#, SQL, React, CSS, Javascript, Bash / Shell, Java, MATLAB, TypeScript, Assembly, Verilog
- **Databases**: Snowflake, PostgreSQL, MySQL, Big Data (Spark, Hadoop)
- **Frameworks**: React (REST APIs), PyTorch, TensorFlow, LangChain, FastAPI, OpenCV, MediaPipe, Pytest
- **Developer Tools**: AWS (DVA-C02), Docker, Kubernetes, Git, MLflow, CI/CD, Valgrind, CMake
- **Libraries**: Pandas, NumPy, Scikit-learn, Tailwind CSS, Node.js, FAISS, Transformers, Matplotlib, Whisper

EXPERIENCE

Applied Machine Learning Intern | Xero May 2024 - August 2025

- Developed an OCR-based ML document understanding system using Python using Tensorflow to extract financial data from over 100k+ documents per day (invoices, receipts, bank statements), achieving 90% accuracy across diverse document layouts.
- Developed scalable cloud pipelines on AWS EC2 for distributed training and inference workflows.
- Containerized and deployed services using Kubernetes, improving reliability and release consistency.
- Orchestrated data and ML workflows using Prefect, spanning ingestion, training, and evaluation.
- Built data processing pipelines with Spark, Hadoop, SQL, and Snowflake for large-scale analytics.
- Implemented CI/CD and A/B testing pipelines to support reliable production model deployment and evaluation.

Research Intern (NSERC Undergraduate Summer Research Awards) | Microfluidics BioMEMS Lab May 2023 - September 2023

- Developed wearable sensors and Python signal processing pipelines to collect and denoise IMU motion data for stroke recovery.
- Developed a PyTorch LSTM model to classify upper extremity movements.

PROJECTS

"Multi-Model AI-Driven Presentation Evaluator" - Capstone Project *(Python, React, Jupyter)*

- Engineered a full-stack system using React and Python to analyze presentation skills, with a user-reported overall presentation skill improvement rate of 93%.
- Integrated open-source tools such as openSMILE, Py-Feat and Whisper to process multi-modal streams of data.

"Genshin Impact RAG System" - Personal Project *(Python, LangChain, FAISS)*

- Developed a RAG pipeline using LangChain and FAISS to eliminate LLM hallucinations for popular video game, Genshin Impact.
- Engineered an automated data ingestion system to chunk unstructured web data into high-dimensional vector embeddings

"The Traveller's Atlas" - Group Project *(C++, API, Multithreading)*

- Developed a high-performance mapping engine using multithreading and custom gridding algorithms, achieving a 600% speedup in data loading and 10x higher frame rates.
- Implemented optimized Dijkstra and A* pathfinding algorithms with OpenStreetMap API integration.

PUBLICATIONS

- **Trust Semantics Distillation for Collaborator Selection via Memory-Augmented Agentic AI**., IEEE Network, 2026
- **Beyond Words – A Multimodal AI Coaching Tool to Enhance Communication Skill Development in Engineering Education**, First Author, Accepted for the 2026 ASEE Annual Conference Exposition, June 2026.

LEADERSHIP

VP Marketing | UofT Machine Intelligence Student Team May 2024 - May 2025

- Scaled community to 3,500+ members and drove a 36% growth in social media following by leveraging analytics to optimize campaign performance and acted as Marketing Chair in GenAI Genesis Hackathon, gathering over 2200 hacker applications.